Home Assignment-2

U Parvathi
2400030589

① Branch & bound method

Max $Z = 5x_1 + 3x_2$

STC's $= 4x_1 + 5x_2 \leq 6$, $x_1 + 3x_2 < 8$,

$2x_2 \leq 8$

$x_1, x_2 \geq 0$ & integer

Concert constraints from $7x_1 \leq 8 \Rightarrow x_2 \leq 10$

$\Rightarrow x_2 = 0$

check feasible integer points

$x_2 = 0$

$4x_1 \leq 6$

$x_1 \leq 15$

$-1x_1 = 0.1$

x_1	x_2	$Z = 5x_1 + 3x_2$
0	0	0
1	0	5

Case 2: $x_2 = 1$

$4x_1 + 3 \leq 6 \Rightarrow 4x_1 \leq 1 \Rightarrow x_1 = 0$

x_1	x_2	Z
0	1	3

Scanned with OKEN Scanner

step 3: optimal solution

$Z_{max} = 5$ at $(x_1, x_2) = (1, 0)$

② Newtons method

S-1 :- $f'(x) = 2x - 6$

S-2 :- second derivative

$f''(x) = 2$

S-3 :- formula

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

S-4 :- Interative $(x_0 = 0)$ $x_1 = 0$ $x_2 = 3$

S-5 :- min value $= 11$ at $x = 3$

③ Given, maximize $Z = x_1 + 4x_2$

STI's $= 6x_1 + 4x_2 \leq 24$

$2x_1 + 2x_2 \leq 6$

S-1: solve LP relation

simply : $x_1 + x_2 \leq 3$

Scanned with OKEN Scanner

x_1	x_2	z
0	3	12
2	1	6

S-2: Integer check best solution

$(x_1, x_2) = (0,3)$

$\boxed{\therefore z_{max} = 12 \text{ at } (0,3)}$

④ Newton's method optimization

steps:-

→ Choose initial points x

→ Compute first derivative $f'(x)$

→ Compute second derivative $f''(x)$

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

example: $f(x) = x^2 - 4x + 5$

$f'(x) = 2x - 4$

$f''(x) = 2$

Scanned with OKEN Scanner

start $x_0 = 0$, $x_1 = 2$

minimum at $x = 2$

⑤ Algorithm steps:

→ solve LP relonation ignore integer

→ check solution

→ If integer → optimal solution

→ If not → get to branching

→ branching → create sub problem

$x \leq [value]$, $x \geq [value]$

→ bounding:

↳ solve each subproblem if

↳ more than current best

↳ integer solution found.

→ Repeat until all nodes explored

Scanned with OKEN Scanner